

HAT-P-36b

R-0925

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-36_230430 · created 2026-07-29T10:15:38+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460064.8963 +/- 0.0045 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.141 +/- 0.016
Transit depth (Rp/Rs)^2	2.0 +/- 0.44 [%]
Orbital Inclination (inc)	80.26 +/- 3.1
Airmass coefficient 1 (a1)	0.966 +/- 0.014
Airmass coefficient 2 (a2)	0.0193 +/- 0.0042
Scatter in the residuals of the lightcurve fit is	1.5 %
Best Comparison Star	#3 - [277, 386]
Optimal Method	PSF photometry
Transit Duration (day)	0.069 +/- 0.024

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.141 ± 0.016 vs 0.126 (deviation 0.87σ)
Duration fit vs published	0.069 d vs 0.0925 d (deviation 0.91σ)
Mid-time O–C	12.5 min
Depth SNR	8.2
Residual scatter	1.5 %

Light curve

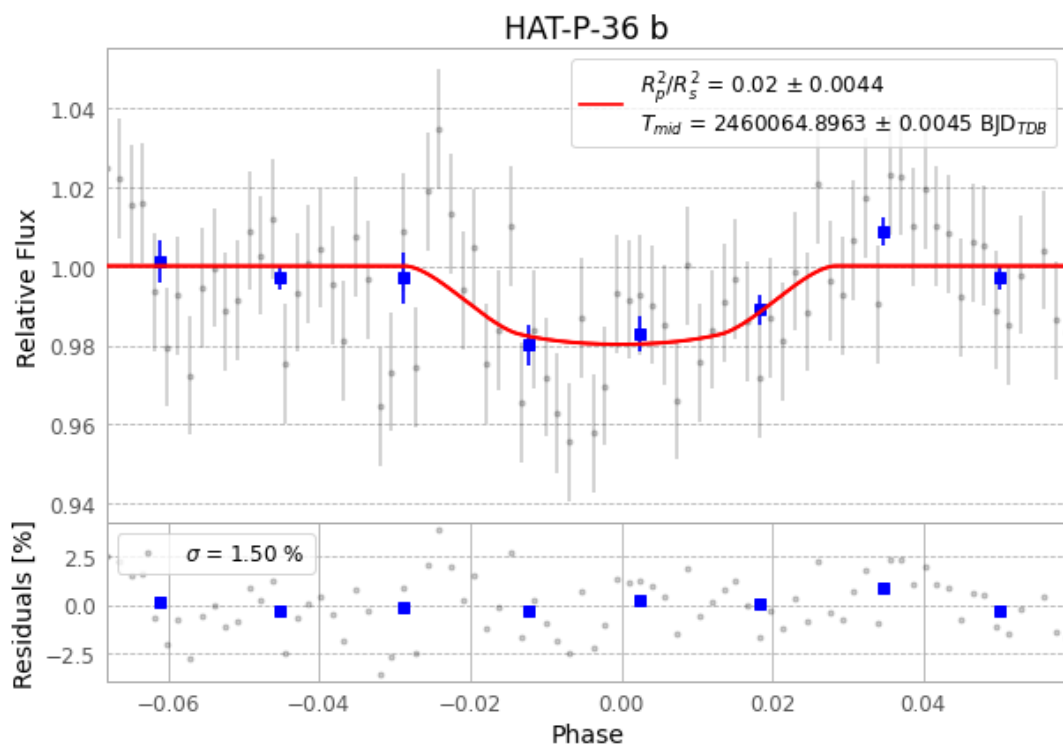


Figure 1 — FinalLightCurve_HAT-P-36 b_2023-04-30.png (SHA-256 093c849bf27b6fd5...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [285, 402], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ad0875ed4f961c42...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

82 FITS frames (54 MB), HATP-36230430071515.FITS → HATP-36230430111815.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-230430.log (075152772bc8...), FinalLightCurve_HAT-P-36 b_2023-04-30.png (093c849bf27b...), FinalLightCurve_HAT-P-36 b_2023-04-30.csv (a7aa3661795a...)

Compute resources

Wall time 125.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 10:15Z.